Homework: the lecture's examples, with both tools

Deadline: **November 17, 2026**. Read the main README first (setup and the two commands). Write your answers in ANSWERS.md.

Exercises 1–4 use OpenJML (way A, loop invariants), exercises 5–7 use JBMC on the same programs (way B, unrolling), and exercise 8 compares them.

OpenJML

1. Absolute

Run `./esc hw/Absolute.java` *before* writing anything.

- **Q** What does OpenJML complain about, and for which input?

Then write the contract of `abs`: the result is non-negative and is `n` or `-n`.

- **Q** Which `requires` did you need, and why? Could you avoid it by changing the `ensures` instead?

2. SumX

The example of the lecture. The contract and an invariant are given.

- **Q** Which proof obligations fail? Is the program wrong?

Strengthen the invariant until the method verifies.

- **Q** The precondition says `x <= 1000`. What is the largest bound *N* for which `x <= N` still verifies, and what does OpenJML report for *N* + *1*? Why?

Then put back `x <= 1000`: exercise 6 uses it.

3. Max

Write loop invariants (and a `decreases` clause) for `max`. Keep going until OpenJML reports only failures you cannot fix with better invariants.

- **Q** Is the remaining failure caused by your invariants or by the code? Write your hypothesis — **do not fix the code yet**: you will check it with JBMC in exercise 5.

4. BinarySearch

The contracts and loop invariants are given.

- **Q** What does OpenJML report, and in which method? Describe a concrete call that triggers it. How large must the array be?

- **Q** `search` is verified using only the *contract* of `midpoint`, not its code. Remove `/*@ pure @*/` from `midpoint`: what happens, and why?

Put `pure` back, fix `midpoint` (keep the original line in a comment: you will need it in exercise 7) and verify.

JBMC

The harnesses `*Check.java` call the code you just verified.

5. Max, again

With `Max.java` as you left it in exercise 3, run `./bmc MaxCheck.check --unwind 3 --no-unwinding-assertions`.

- **Q** What input did JBMC find, and what goes wrong with it? Was your hypothesis of exercise 3 right?
- **Q** What are the smallest `--unwind` and the smallest `--max-nondet-array-length` that still find the bug (keep `--no-unwinding-assertions`)? Relate them to the lecture's small scope hypothesis.

Now fix `Max.max`, and check it with both tools.

- **Q** Run `./bmc MaxCheck.check --unwind 5` (with unwinding assertions). Why does it fail now? Add `--max-nondet-array-length 4`: what is the smallest `--unwind` that gives VERIFICATION SUCCESSFUL, and why not one less? What exactly has been proved?

6. SumX, again

Write the harness `SumXCheck.check`, with the contract of exercise 2.

- **Q** `--unwind 10 --no-unwinding-assertions` says SUCCESSFUL. Which values of `x` were really checked? Drop `--no-unwinding-assertions`: why does it fail? Which *K* turns the result into a proof? Try it and compare the formula size and time for K = 10, 100 and that *K*.
- **Q** What did OpenJML need from you to prove `sum` correct, and what did JBMC need?

7. BinarySearch, again

Put back the original `midpoint` (`(lo + hi) / 2`) and run `./bmc BinarySearchCheck.search --unwind 6 --max-nondet-array-length 5`.

- **Q** It says SUCCESSFUL, but OpenJML found a bug. Why does JBMC miss it with this harness? Could a larger bound help in practice?

Write the harness `BinarySearchCheck.midpoint` (the contract of `midpoint`) and run it with `--unwind 1`.

- **Q** What did it find?

Restore your fix and check again.

Comparison

8. Comparison

Fill in the table in ANSWERS.md for `SumX`, `Max` and `BinarySearch`.

- **Q** When would you use each tool?

Before you submit

- `./esc` says OK for every file in `hw/`: `Absolute`, `SumX` (with `x <= 1000`), `Max` (invariants and the fix) and `BinarySearch` (the fixed `midpoint`).
- `./bmc` gives VERIFICATION SUCCESSFUL for `MaxCheck.check --unwind 5 --max-nondet-array-length 4`, `SumXCheck.check --unwind 1001` and `BinarySearchCheck.midpoint --unwind 1`.
- No `assert false;  // remove this line` is left in a harness.
- Every question in `ANSWERS.md` is answered.